

restore limited level of resource redundancy while adaptively adjust regional resilience with lower investment and operational cost.

4.1 Introduction

Supply chains have recently experienced tremendous stresses due to disruptive weather, supplier disruptions, trade wars and other geopolitical events, and recent flus and pandemics. For example, demand for reagents for developing and manufacturing COVID19 testing kits and vaccines have reduced the availability of those same reagents needed for, for example, cancer immunotherapy manufacturing. Such stresses, and others described in (Henriksen & Selwyn, 2020; Skibsted, 2020; Lin & Lanng, 2020) and elsewhere, have revealed the fragility of the modern supply chain (especially supply chains designed to be lean), resulting in a heightened interest in supply chain resilience and agility – the ability to quickly adjust to sudden disruptive changes that can negatively affect supply chain performance as it affects the firm’s objectives (e.g., shareholder value) and in times of national crises, as it more broadly affects the Nation’s health and economy. Supply chain resilience also can be strengthened by, for example, increasing the inventory levels of raw material, work-in-progress, and the final product, adding manufacturing and/or storage capacity to improve mitigating potential supplier disruption. However, such procedures for strengthening supply chain resilience can be expensive. Competitive advantage will result if a firm’s supply chain resilience and agility is identical to the competition’s but at lower cost.

Data-driven rebalancing of inventories by transshipment of raw materials, work-in-progress, and finished product can be part of the solution. Supply chain resilience can be further enhanced by also relocating portable, modular production capacity (e.g., small 3D printers, bioreactors) and/or storage capacity (e.g., smart lockers) based on real-time data-driven demand analysis (Dolgui, 2020; Maladi, 2019; Faugere, 2020). The result is supply chain performance that without customer service level reduction blends the advantages of distributed supply chain systems (having inventory and/or manufacturing capacity

close to demand to enable fast fulfillment) and centralized supply chain systems (to enable economies of scale, inventory and risk ‘pooling’, reduced total safety inventory, and reduced total capital expenditures). Lean supply chains that minimize cost may be unable to effectively respond to and recover from unexpected and disruptive change (see [4]). Therefore, a future supply chain is motivated to respond to and recover from them by increasing manufacturing capacity where needed in the network for enhanced resilience and post-disruption surge and then reducing manufacturing capacity once recovery has been achieved. We define such system to be dynamically resilient, which is either lean or agile, depending on need.

In this chapter, we investigate several key fundamental research challenges for jointly making reusable manufacturing equipment investment and consumable inventory decisions in decentralized manufacturing networks for build-to-order (BTO) product subject to potential supply disruptions. Our research is motivated by the emerging autologous regenerative medicine industry, where a unit of manufacturing capacity is a bioreactor, the consumable is a reagent, and the supplier base is nascent (e.g., key reagents may have only a single supplier source) and hence is a source of significant risk. However, the models and analysis proposed can be extended to any BTO manufacturing-fulfillment network where capacity planning for both reusable and consumable resources must be made in order to reduce total manufacturing and logistic costs, and ensure system resilience against supplier-based risks. More specifically, given a description of the supply chain network and of the supply risk, we propose analytical and numerical methods for determining (i) the right level of manufacturing capacity and an optimal (or near optimal) replenishment policy for the reagent at each manufacturing facility and (ii) good transshipment and manufacturing capacity relocation policies order to ensure competitiveness and resilience.

4.1.1 Chapter Organization

We discuss related literature in Section 4.2. Adopting the modeling approach for supplier disruption risk in Chapter 3, we model two decentralized network design problem in Section 4.3. Analysis of the network models are presented in Section 4.4. We propose four heuristic algorithms to solve the proposed models in Section 4.5, including two stochastic approximation heuristics and two artificial intelligence (AI) based heuristics. In Section 4.6, we use a case study to demonstrate the potential of proposed models/algorithms in increasing supply chain resilience while maintaining cost efficiency. We conclude this chapter in Section 4.7.

4.2 Literature Review

This is a place holder for literature review.

4.3 CP in a Decentralized Facility Network

We consider modeling a decentralized facility network with multiple facilities at different locations. One possible design is an Isolatedly operated manufacturing/service Network (Iso-Net) where each facility is operated separately processing its regional demand with its own resources. Another possible design is a Coordinately operated manufacturing/service Network (Co-Net) that all facilities are able to collaborate with others if needed.

We consider a finite horizon capacity planning problem and assume the total number of bioreactors is fixed throughout the planning horizon.

- In an Iso-Net, the decisions/controls include
 - for bioreactors
 - (a) investment plan: determine the total number of bioreactors in the network
 - (b) assignment plan: assign bioreactors to facilities in the network.

- for reagent: replenishment plan: determine recurrent inventory replenishment quantities.
- In a Co-Net, the decisions/controls include
 - for specimen: demand sharing plan: determine specimen recurrent transshipment quantities.
 - for bioreactors:
 - (a) investment plan: determine the total number of bioreactors in the network
 - (b) assignment plan: assign bioreactors to facilities in the network
 - (c) equipment relocation plan: determine recurrent bioreactor relocation quantities
 - for reagent:
 - (a) replenishment plan: determine recurrent inventory replenishment quantities
 - (b) material sharing plan: determine recurrent reagent transshipment quantities.

We consider the determination of bioreactor investment and assignment plans as the **CP-design problem**, and consider the determination of demand sharing, bioreactor equipment relocation, reagent replenishment and material sharing plans as the *CP-operation problem*.

The indices of the models, discussed in this chapter, includes (i) location $l \in [L]$, where L is the number of locations, and $[\cdot]$ is a set operator such that $[z] = \{1, \dots, z\}$ for $z \in \mathbb{Z}^+$, (ii) decision epoch $t \in [T]$, and (iii) bioreactor state $\tau \in \{0, 1, \dots, T - 1\}$, where τ is the number of periods needed to produce a therapy. At epoch t , let

- $s_t^l \geq 0$ be the number of specimens in facility l , waiting for therapy manufacturing to begin. s_t is the collection of queue lengths at all facilities;

- $\mathbf{b}_t^l = (b_t^{l,0}, \dots, b_t^{l,T-1})$ is the vector of bioreactors at different production stages at facility l , where $b_t^{l,\tau} \geq 0$ bioreactors will complete their manufacturing processes in τ epochs, and hence $b_t^{l,0}$ is the current number of idle bioreactors. $\mathbf{b}_t$ is the collection of bioreactor status vectors at all facilities;
- $r_t^l \geq 0$ is the number of units of reagent at facility l . $\mathbf{r}_t$ is the collection of reagent inventory at all facilities;
- $A_t^l \geq 0$ is the maximum units of reagent that can be supplied to facility l . Let $\mathbf{A}_t$ be the collection of supplier capacities at all facilities.

If we assume the dynamic process of $\{\mathbf{A}_t\}$ is Markovian, the facility l at epoch t can be described as $x_t^l = (s_t^l, \mathbf{b}_t^l, r_t^l, A_t^l)$, and $\mathbf{x}_t = (\mathbf{s}_t, \mathbf{b}_t, \mathbf{r}_t, \mathbf{A}_t)$ is the system state. We remark that the dynamic process of $\{\mathbf{A}_t\}$ may also be models as a multiple time series model. Let $\mathcal{H}_t^l$ be a historical time series of values of A_t^l , and $\mathcal{H}_t$ be a collection of time series $\mathcal{H}_t^l$ for all l . Then the facility l at epoch t should be described as $x_t^l = (s_t^l, \mathbf{b}_t^l, r_t^l, \mathcal{H}_t^l)$ in order to adopt a Markov Decision Process modeling approach. Therefore, a system state at epoch t is $\mathbf{x}_t = (\mathbf{s}_t, \mathbf{b}_t, \mathbf{r}_t, \mathcal{H}_t)$. If the time series model is trained offline, we then only need to carry over a segment of $\mathcal{H}_t$, e.g. the latest k supplier capacity historical data (i.e. $\mathbf{A}_{t-k+1}, \dots, \mathbf{A}_t$), to produce forecasts of $\mathbf{A}_{t+1}$. It is easy to see that the Markovian case is a special case of $k = 1$.

Given a bioreactor investment plan (determination of bioreactor quantities at each location), there are five sets of actions at each decision epoch $t \in [\mathcal{T}]$:

- $m_t^l \in \{0\} \cup [\min\{s_t^l, b_t^{l,0}, r_t^l\}]$ is the number of productions started at epoch t
- $w_t^l \in \mathbb{Z}$ is the number of specimens transshipped into location l after m_t^l is decided and before epoch $t + 1$. A negative valued w_t^l indicates transshipping specimens out from facility l . Hence a balance of total transshipment suggests that $\sum_l w_t^l = 0$

- $a_t^l \in \{0\} \cup [A_t^l]$ is the number of units of reagent ordered at epoch t and received before epoch $t + 1$
- $e_t^l \in \mathbb{Z}$ is the number of units of reagent transshipment after a_t is received and before epoch $t + 1$. Similar with specimen transshipping, a balance of total transshipment suggests that $\sum_l e_t^l = 0$
- $q_t^l \in \mathbb{Z}$ is the number of bioreactor relocation after $b_t^{l,1}$ is released to idle and before epoch $t + 1$. Similar with specimen transshipping, a balance of total relocation suggests that $\sum_l q_t^l = 0$.

Let $\mathbf{d}_t = (d_t^1, \dots, d_t^L)$ be stochastic demands occurred during period t , the system dynamics can be represented as

$$\begin{aligned}
 s_{t+1}^l &= s_t^l - m_t^l + d_t^l + w_t^l, \\
 b_{t+1}^{l,\tau} &= \begin{cases} b_t^{l,0} - m_t^l + b_t^{l,1} + q_t^l & \tau = 0, \\ b_t^{l,\tau+1} & 1 \leq \tau \leq T-2, \\ m_t^l & \tau = T-1, \end{cases} \\
 r_{t+1}^l &= r_t^l - m_t^l + a_t^l, \\
 \mathcal{H}_{t+1}^l &= \mathcal{F}^l(\mathcal{H}_t)
 \end{aligned}$$

where $\mathcal{F}^l$ is a probabilistic mapping that maps historical trajectories $\mathcal{H}_t$ to an updated supplier capacity trajectory $\mathcal{H}_{t+1}^l = \mathcal{H}_t^l \cup \{A_{t+1}^l\}$ with conditional probability $P(A_{t+1}^l | \mathcal{H}_t)$. In a Markov process case, i.e. $\mathcal{H}_t = \mathbf{A}_t$, $\mathcal{F}^l$ is a probabilistic mapping with transition probability $P(A_{t+1}^l | \mathbf{A}_t)$. We assume from this point that supplier disruption processes can be modeled as exogenous Markov processes, i.e. $\mathbf{x}_t = (\mathbf{s}_t, \mathbf{b}_t, \mathbf{r}_t, \mathbf{A}_t)$, and all results can be easily extended to models with general transition function $\mathcal{F}$.

We assume that facility l is able to transship at most $s_t^l - m_t^l + d_t^l$ specimens or receive

at most $\sum_{i \neq l} (s_t^i - m_t^i + d_t^i)$ specimens in period t , so that

$$-(s_t^l - m_t^l + d_t^l) \leq w_t^l \leq \sum_{i \neq l} (s_t^i - m_t^i + d_t^i).$$

Given the fact that $\sum_l w_t^l = 0$, the right-hand side inequality is redundant and thus the action space of specimen transshipment is

$$\{\mathbf{w}_t \in \mathbb{Z}^L : w_t^l \geq -(s_t^l - m_t^l + d_t^l), \sum_l w_t^l = 0\}.$$

Similarly, the action space of reagent transshipment and bioreactor relocation are,

$$\{\mathbf{e}_t \in \mathbb{Z}^L : e_t \geq -(r_t^l - m_t^l + a_t^l), \sum_l e_t^l = 0\}$$

and

$$\{\mathbf{q}_t \in \mathbb{Z}^L : q_t^l \geq -(b_t^{l,0} - m_t^l + b_t^{l,1}), \sum_l q_t^l = 0\}$$

respectively.

Let the single period cost accrued in period t be additive over facilities

$$C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) = \sum_l C^l(w_t^l, a_t^l, e_t^l, q_t^l, m_t^l, d_t^l | x_t^l),$$

and each single facility cost, i.e. $C^l(w_t^l, a_t^l, e_t^l, q_t^l, m_t^l, d_t^l | x_t^l)$, consist multiple cost components

- $c_R a_t^l$ is the reagent replenishment cost for the reagent, where c_R is the cost per unit of reagent
- $h_R (r_{t+1}^l - s_{t+1}^l)^+$ is the reagent overstock holding cost, charged if there are more units of reagent than specimens at epoch $t + 1$, where h_R is the holding cost per excess unit of reagent

- $h_B(b_{t+1}^{l,0} - s_{t+1}^l)^+$ is the bioreactor overstock holding cost, charged if there are more idle bioreactors than specimens at epoch $t + 1$, where h_B is the bioreactor holding cost per excess idle bioreactor
- $p_R(s_{t+1}^l - r_{t+1}^l)^+$ is the reagent understock penalty, charged if there are less units of reagent than specimens at epoch $t + 1$, where p_R is the penalty cost per insufficient unit of reagent
- $p_B(s_{t+1}^l - b_{t+1}^{l,0})^+$ is the bioreactor understock penalty, charged if there are less idle bioreactors than specimens at epoch $t + 1$, where p_B is the penalty cost per insufficient idle bioreactor
- $K_S \frac{|w_t^l|}{2}$ is the cost of transshipping w_t^l specimens from/to facility l that is charged on facility l
- $K_R \frac{|e_t^l|}{2}$ is the cost of transshipping e_t^l units of reagent from/to facility l that is charged on facility l
- $K_B \frac{|q_t^l|}{2}$ is the cost of relocating q_t^l bioreactors from/to facility l that is charged on facility l .

Therefore,

$$C^l(w_t^l, a_t^l, e_t^l, q_t^l, m_t^l, d_t^l | x_t^l) = c_R a_t^l + K_R \frac{|e_t^l|}{2} + h_R(r_{t+1}^l - s_{t+1}^l)^+ + p_R(s_{t+1}^l - r_{t+1}^l)^+ \\ + K_B \frac{|q_t^l|}{2} + h_B(b_{t+1}^{l,0} - s_{t+1}^l)^+ + p_B(s_{t+1}^l - b_{t+1}^{l,0})^+ + K_S \frac{|w_t^l|}{2}$$

Then, the *CP-operation problem* of Co-Net can be modeled as a dynamic programming

problem

$$v_t^*(\mathbf{x}_t) = \min_{\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t \in \mathbb{Z}^L} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) + \beta v_{t+1}^*(\mathbf{x}_{t+1})] \quad (4.1a)$$

$$s.t. \quad 0 \leq \mathbf{a}_t \leq \mathbf{A}_t, \quad (4.1b)$$

$$\sum_l w_t^l = 0, \sum_l e_t^l = 0, \sum_l q_t^l = 0, \quad (4.1c)$$

$$w_t^l \geq -(s_t^l - m_t^l + d_t^l) \quad \forall l, \quad (4.1d)$$

$$e_t^l \geq -(r_t^l - m_t^l + a_t^l) \quad \forall l, \quad (4.1e)$$

$$q_t^l \geq -(b_t^{l,0} - m_t^l + b_t^{l,1}) \quad \forall l, \quad (4.1f)$$

$$0 \leq m_t^l \leq \min\{s_t^l, b_t^{l,0}, r_t^l\} \quad \forall l, \quad (4.1g)$$

and boundary condition $v_{T+1}(x) = 0$. Therefore, the *CP-design problem* of Co-Net model is

$$\min_{b_1^{1,0}, \dots, b_1^{L,0} \in \mathbb{N}} c_B \sum_{l \in [L]} b_1^{l,0} + v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0})) \quad (4.2)$$

We remark that by setting $\mathbf{w}_t = \mathbf{e}_t = \mathbf{q}_t = 0$ for all t in (4.1), the Co-Net model becomes an Iso-Net model, whose *CP-operation problem* is

$$v_t^\dagger(\mathbf{x}_t) = \min_{\mathbf{a}_t, \mathbf{m}_t \in \mathbb{Z}^L} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) + \beta v_{t+1}^\dagger(\mathbf{x}_{t+1})] \quad (4.3)$$

$$s.t. \quad \mathbf{w}_t = \mathbf{e}_t = \mathbf{q}_t = 0,$$

$$0 \leq \mathbf{a}_t \leq \mathbf{A}_t,$$

$$0 \leq m_t^l \leq \min\{s_t^l, b_t^{l,0}, r_t^l\} \quad \forall l.$$

By replacing v_0^* with $v_0^\dagger$, we obtain its *CP-design problem* as

$$\min_{b_1^{1,0}, \dots, b_1^{L,0} \in \mathbb{N}} c_B \sum_{l \in [L]} b_1^{l,0} + v_1^\dagger(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0})). \quad (4.4)$$

4.4 Analysis

In this section, we discuss analytical results of the proposed CP models.

4.4.1 Effects of transshipping/relocation costs

We first investigate the how transshipping/relocation costs, i.e. K_S , K_R and K_B , effect CP-operation decisions. Given a system state $\mathbf{x}_t$, larger transshipping/relocation costs will result in less transshipment or relocation, i.e. $|e_t^l|$, $|q_t^l|$ and $|w_t^l|$, see Theorem 4.4.1

Theorem 4.4.1. *Given a system state $\mathbf{x}_t$, and two CP-operation problem instances whose transshipping/relocation costs are $\{K_S, K_R, K_B\}$ and $\{\bar{K}_S, \bar{K}_R, \bar{K}_B\}$ where*

$$K_S \leq \bar{K}_S, \quad K_R \leq \bar{K}_R, \quad K_B \leq \bar{K}_B,$$

let $\pi^(\mathbf{x}_t)$ and $\bar{\pi}^*(\mathbf{x}_t)$ be optimal policies of the two instances, respectively, i.e.*

$$\begin{aligned} \pi^*(\mathbf{x}_t) &= (\mathbf{w}_t^*(\mathbf{x}_t), \mathbf{a}_t^*(\mathbf{x}_t), \mathbf{e}_t^*(\mathbf{x}_t), \mathbf{q}_t^*(\mathbf{x}_t), \mathbf{m}_t^*(\mathbf{x}_t)) \\ \bar{\pi}^*(\mathbf{x}_t) &= (\bar{\mathbf{w}}_t^*(\mathbf{x}_t), \bar{\mathbf{a}}_t^*(\mathbf{x}_t), \bar{\mathbf{e}}_t^*(\mathbf{x}_t), \bar{\mathbf{q}}_t^*(\mathbf{x}_t), \bar{\mathbf{m}}_t^*(\mathbf{x}_t)). \end{aligned}$$

Then the following inequalities hold component-wisely

$$|\mathbf{e}_t^*(\mathbf{x}_t)| \geq |\bar{\mathbf{e}}_t^*(\mathbf{x}_t)|, \quad |\mathbf{q}_t^*(\mathbf{x}_t)| \geq |\bar{\mathbf{q}}_t^*(\mathbf{x}_t)|, \quad |\mathbf{w}_t^*(\mathbf{x}_t)| \geq |\bar{\mathbf{w}}_t^*(\mathbf{x}_t)|.$$

Proof of Theorem 4.4.1 is included in the Appendix A. We remark that as K_S , K_R and K_B become significantly large, CP-operation problems in a Co-Net will reduce to an Iso-Net, as the huge transshipping and relocation penalties would essentially forbid any sharing of demand, equipment or materials.

Corollary 4.4.2. *Let*

$$\pi^*(\mathbf{x}_t) = (\mathbf{w}_t^*(\mathbf{x}_t), \mathbf{a}_t^*(\mathbf{x}_t), \mathbf{e}_t^*(\mathbf{x}_t), \mathbf{q}_t^*(\mathbf{x}_t), \mathbf{m}_t^*(\mathbf{x}_t))$$

and

$$\pi^\dagger(\mathbf{x}_t) = (\mathbf{0}, \mathbf{a}_t^\dagger(\mathbf{x}_t), \mathbf{0}, \mathbf{0}, \mathbf{m}_t^\dagger(\mathbf{x}_t))$$

be optimal policies of the CP-operation problem of Co-Net and Iso-Net models, respectively, given a same system state $\mathbf{x}_t$. Then $\pi^(\mathbf{x}_t) \rightarrow \pi^\dagger(\mathbf{x}_t)$ as K_S , K_R and K_B go to infinite.*

4.4.2 Bounds on the expected total cost of a Co-Net

Next, we discuss bound results on the expected total cost of a Co-Net. Let $(b_1^{1,0*}, \dots, b_1^{L,0*})$ be an optimal solution for Co-Net CP-design problem (4.2) and the optimal objective value is

$$u^*(b_1^{1,0*}, \dots, b_1^{L,0*}) = c_B \sum_{l \in [L]} b_1^{l,0*} + v_1^*(\mathbf{x}_1(b_1^{1,0*}, \dots, b_1^{L,0*})).$$

Let $(b_1^{1,0\dagger}, \dots, b_1^{L,0\dagger})$ be an optimal solution for Iso-Net CP-design problem (4.4) and the optimal objective value is

$$u^\dagger(b_1^{1,0\dagger}, \dots, b_1^{L,0\dagger}) = c_B \sum_{l \in [L]} b_1^{l,0\dagger} + v_1^\dagger(\mathbf{x}_1(b_1^{1,0\dagger}, \dots, b_1^{L,0\dagger})).$$

Theorem 4.4.3. *The Co-Net model is a relaxation of Iso-Net model, and hence*

$$u^*(b_1^{1,0*}, \dots, b_1^{L,0*}) \leq u^\dagger(b_1^{1,0\dagger}, \dots, b_1^{L,0\dagger}).$$

We have shown that the Iso-Net produces an upper bound on the optimal expected total cost of the Co-Net, i.e. $u^*(b_1^{1,0*}, \dots, b_1^{L,0*})$. We further construct a lower bound on $u^*(b_1^{1,0*}, \dots, b_1^{L,0*})$. The lower bound construction include two parts

- set $K_S = K_R = K_B = 0$
- allow immediate demand/equipment/material sharing at the beginning of epoch t .

Mathematically, let $a_t = \sum_{l \in [L]} a_t^l$, $s_t = \sum_{l \in [L]} s_t^l$, $b_t^\tau = \sum_{l \in [L]} b_t^{l,\tau}$ and $r_t = \sum_{l \in [L]} r_t^l$, $m_t = \sum_{l \in [L]} m_t^l$, $d_t = \sum_{l \in [L]} d_t^l$ for all t and τ . Construct a single period cost

$$\begin{aligned} C^\#(a_t, d_t | \mathbf{x}_t) = & c_R a_t + h_R(r_t + a_t - s_t - d_t)^+ + p_R(s_t + d_t - r_t - a_t)^+ \\ & + h_B(b_t^0 + b_t^1 - s_t - d_t)^+ + p_B(s_t + d_t - b_t^0 - b_t^1)^+, \end{aligned}$$

and revised system state $x_t = (s_t, b_t, r_t, \mathbf{A}_t)$, where $b_t = \{b_t^\tau\}$. Then we obtain a dynamic programming (DP) model which is a single facility CP-operation problem

$$v_t^\#(x_t) = \min_{a_t, m_t \in \mathbb{Z}} \mathbb{E}[C^\#(a_t, d_t | x_t) + \beta v_{t+1}^\#(x_{t+1})] \quad (4.5)$$

$$s.t. \quad 0 \leq a_t \leq A_t,$$

$$0 \leq m_t \leq \min \{s_t, b_t^0, r_t\},$$

and the value of A_{t+1} in the upcoming epoch follows a convoluted distribution $P(\sum_{l \in [L]} A_{t+1}^l | \mathbf{A}_t)$.

And a single facility CP-design problem is formulated as

$$\min_{b_1^0 \in \mathbb{N}} c_B b_1^0 + v_1^\#(x_1(b_1^0)). \quad (4.6)$$

We name the reconstructed CP problem Single Location Approximation (L1-Apx). Let $b_1^{0\#}$ be an optimal solution of (4.6) and let

$$u^\#(b_1^{0\#}) = c_B b_1^{0\#} + v_1^\#(x_1(b_1^{0\#}))$$

be its optimal objective value.

Theorem 4.4.4. *The L1-Apx model is a relaxation of Co-Net, and hence*

$$u^\#(b_1^{0\#}) \leq u^*(b_1^{1,0*}, \dots, b_1^{L,0*}).$$

Theorems 4.4.3 and 4.4.4 suggest an inequality chain

$$u^\#(b_1^{0\#}) \leq u^*(b_1^{1,0*}, \dots, b_1^{L,0*}) \leq u^\dagger(b_1^{1,0\dagger}, \dots, b_1^{L,0\dagger}),$$

and the upper and lower bound can be solved with the help of $L = 1$ model discussed extensively in Chapter 3 as the FBQ-sdr model.

- The L1-Apx is, essentially, an FBQ-sdr model with slight modification on the dynamic of A_t
- The Iso-Net is separable by facility, i.e.

$$\begin{aligned} v_t^l(x_t^l) &= \min_{a_t^l, m_t^l \in \mathbb{Z}} \mathbb{E}[C^l(w_t^l, a_t^l, e_t^l, q_t^l, d_t^l | x_t^l) + \beta v_{t+1}^l(x_{t+1}^l)] \\ \text{s.t. } & 0 \leq a_t^l \leq A_t^l \\ & 0 \leq m_t^l \leq \min\{s_t^l, b_t^{l,0}, r_t^l\}, \end{aligned}$$

and $v_1(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0})) = \sum_l v_1^l(x_1^l(b_1^{l,0}))$, where $x_1^l(b_1^{l,0})$ is the starting state of facility l which only depends on $b_1^{l,0}$. Therefore, it is equivalent to solve L sperated FBQ-sdr models.

4.4.3 Closed-form solution for $\mathbf{m}_t$

Theorem 4.4.5. *Given a system state $\mathbf{x}_t$, there exist closed-form solutions for $\mathbf{m}_t^*(\mathbf{x}_t)$ for all l ,*

$$m_t^{l*}(x_t^l) = \min\{s_t^l, b_t^{l,0}, r_t^l\}.$$

Proof of Theorem 4.4.5 is included in the Appendix A. The proof contains two parts:

(i) we first show that if there exist some $m_t^l < \min\{s_t^l, b_t^{l,0}, r_t^l\}$, we can always find a larger feasible m_t^l that gives no higher single period cost; (ii) we next show that inductively the same fact hold for multi-period cost-to-go function. Therefore, we reduce the action space by transforming decision variables m_t^l to deterministic variables. Managerial insight of Theorem 4.4.5 is that each facility should process at most of its own pending productions and only consider transshipping slack materials and equipment. For the Iso-Net model, $m_t^{\dagger}(x_t^l) = \min\{s_t^l, b_t^{l,0}, r_t^l\}$ is guaranteed by the results in Chapter 3. We let $m_t^l = \min\{s_t^l, b_t^{l,0}, r_t^l\}$ from this point of the chapter.

4.4.4 Convexity of CP-design problems

Let $u^*(b_1^{1,0}, \dots, b_1^{L,0}) = c_B \sum_l b_1^{l,0} + v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ be the Co-Net expected total cost given a system design $(b_1^{1,0}, \dots, b_1^{L,0})$.

Theorem 4.4.6. $u^*(b_1^{1,0}, \dots, b_1^{L,0}) = c_B \sum_l b_1^{l,0} + v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ is convex in design variables $(b_1^{1,0}, \dots, b_1^{L,0})$, and hence the CP-design problem (4.2) is a convex optimization problem.

Theorem 4.4.6 suggest that a gradient search algorithm can be used to find a global optimal of Co-Net master problem 4.2 with optimality and convergence guarantees (see results in [43]). The proof of Theorem 4.4.6 is included in Appendix A. Let $u^{\dagger}(b_1^{1,0}, \dots, b_1^{L,0}) = c_B \sum_l b_1^{l,0} + v_1^{\dagger}(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ be the Iso-Net expected total cost given a system design $(b_1^{1,0}, \dots, b_1^{L,0})$, the following corollary is straightforward.

Corollary 4.4.7. $u^{\dagger}(b_1^{1,0}, \dots, b_1^{L,0}) = c_B \sum_l b_1^{l,0} + v_1^{\dagger}(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ is convex in design variables $(b_1^{1,0}, \dots, b_1^{L,0})$, and hence the CP-design problem (4.4) is a convex optimization problem.

4.5 Algorithm Design

In this section, we introduce solution algorithms for the Co-Net model, i.e. CP-operation problem (4.1) and CP-design problem (4.2). Solving the Iso-Net model, i.e. CP-operation problem (4.3) and CP-design problem (4.4), is equivalent to solve L FBQ-sdr problems, whose algorithms are introduced in Chapter 3.

4.5.1 Solving Co-Net CP-design problem

Suppose that given a bioreactor investment and assignment plan, i.e. $b_1^{1,0}, \dots, b_1^{L,0}$, we are able to evaluate $v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$. The CP-design problem (4.2) can be solved using an Iterative Coordinate Descent (ICD) algorithm. Optimality and convergence of the coordinate descent algorithm is guaranteed by Theorem 4.4.6.

While a stopping criterion (e.g. maximum iteration, threshold cost improvement and etc.) is not met in iteration k , the ICD algorithm iteratively sample a coordinate $i_k \in [L]$, choose a searching step size δ_k , and evaluate

$$v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{i_k,0} - \delta_k, \dots, b_1^{L,0})) \quad \text{and} \quad v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{i_k,0} + \delta_k, \dots, b_1^{L,0})).$$

δ_k is a non-increasing integer searching step size such that $\lim_{k \rightarrow \infty} \delta_k = 1$. In the next iteration, update the value of $b_t^{i_k,0}$ to either $b_t^{i_k,0} - \delta_k$ or $b_t^{i_k,0} + \delta_k$ who produces a smaller total cost $u^*(\cdot)$. Otherwise, keep the current bioreactor investment plan, sample the next search coordinate i_{k+1} and repeat the evaluation and updating procedures. Details of ICD is summarized in Algorithm 3.

We note when choosing a maximum iteration stopping criteria, it is practical to choose $K = zL$ where $z \in \mathbb{N}$ is the number of master iterations. In each master iteration for sampling i_k , we create a random permutation sequence of L facilities, so that all coordinates are evaluated in each master iteration. We also remark that, the searching step size is fixed in each master iteration and diminished in the next master iteration. A practical example

Algorithm 3: ICD algorithm for solving (4.2)

```

1 Initialization: choose an initial bioreactor investment and assignment plan
   $(b_1^{1,0}, \dots, b_1^{L,0})$ ; choose an initial searching step size  $\delta_1$ ; choose maximum iteration
   $K$ , set  $k \leftarrow 1$ ;
2 Evaluate  $v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ , and calculate  $u^*(b_1^{1,0}, \dots, b_1^{L,0})$ ;
3 while  $k \leq K$  do
4   Sample an improving coordinate  $i_k \in [L]$ ;
5   Evaluate  $v_1^*(\mathbf{x}_t(b_1^{1,0}, \dots, b_1^{i_k,0} - \delta_k, \dots, b_1^{L,0}))$  and
      $v_1^*(\mathbf{x}_t(b_1^{1,0}, \dots, b_1^{i_k,0} + \delta_k, \dots, b_1^{L,0}))$ ;
6   Calculate  $u^*(b_1^{1,0}, \dots, b_1^{i_k,0} - \delta_k, \dots, b_1^{L,0})$  and  $u^*(b_1^{1,0}, \dots, b_1^{i_k,0} + \delta_k, \dots, b_1^{L,0})$ ;
7   if  $u^*(b_1^{1,0}, \dots, b_1^{i_k,0} - \delta_k, \dots, b_1^{L,0}) < u^*(b_1^{1,0}, \dots, b_1^{i_k,0}, \dots, b_1^{L,0})$  then
8      $b_t^{i_k,0} \leftarrow b_t^{i_k,0} - \delta_k$ ;
9   else if  $u^*(b_1^{1,0}, \dots, b_1^{i_k,0} + \delta_k, \dots, b_1^{L,0}) < u^*(b_1^{1,0}, \dots, b_1^{i_k,0}, \dots, b_1^{L,0})$  then
10     $b_t^{i_k,0} \leftarrow b_t^{i_k,0} + \delta_k$ 
11  end
12  Set  $k \leftarrow k + 1$  and update searching step size  $\delta_k$ ;
13 end
14 Output: Bioreactor investment and assignment plan  $(b_1^{1,0}, \dots, b_1^{L,0})$  and expected
    total cost  $u^*(b_1^{1,0}, \dots, b_1^{L,0})$ .

```

for choosing a search step size is to set $\delta_k = \lceil \frac{\delta_0}{\lceil k/L \rceil} \rceil$. It remains to show how to evaluate $v_1^*(\mathbf{x}_1(b_1^{1,0}, \dots, b_1^{L,0}))$ to complete the Co-Net solution algorithm.

4.5.2 Solving Co-Net CP-operation problem

Given a bioreactor investment and assignment plan, i.e. $b_1^{1,0}, \dots, b_1^{L,0}$, a problem instance of CP-operation is formulated. It is challenge to solve an exact DP problem (4.1) due to

- state space complexity
- stochasticity of demand process and supplier disruption process at all locations.

The output of solving CP-operation problem is a policy which is a functional mapping a system state to a control action. We present several near-optimal heuristics which approximately solve the DP by solving surrogate models which have approximated cost-to-go functions, i.e. $v_{t+1}^*(\mathbf{x}_t)$. Let $\pi = \{\pi_t\}_{t \in [\mathcal{T}]}$ be a CP operation policy, and $v_1^\pi(\mathbf{x}_t(b_1^{1,0}, \dots, b_1^{L,0}))$ is the expected total operation cost when policy π is implemented. We approximate $v_1^*(\mathbf{x}_t(b_1^{1,0}, \dots, b_1^{L,0}))$

with $v_1^\pi(\mathbf{x}_t(b_1^{1,0}, \dots, b_1^{L,0}))$ when implementing the ICD algorithm (Algorithm 3 and hence we can obtain a solution for the CP-design problem. The proposed ADP methods belong to *cost approximation* approaches in [44] with near-optimal performance guarantees. For the ease of demonstration, let $\Pi(\mathbf{x}_t)$ be the action space at epoch t , i.e.

$$\begin{aligned} \Pi(\mathbf{x}_t) = \{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \mathbb{Z}^{4L} : 0 \leq \mathbf{a}_t \leq \mathbf{A}_t, \sum_l w_t^l = 0, \sum_l e_t^l = 0, \sum_l q_t^l = 0, \\ w_t^l \geq -(s_t^l - m_t^l + d_t^l), e_t^l \geq -(r_t^l - m_t^l + a_t^l), q_t^l \geq -(b_t^{l,0} - m_t^l + b_t^{l,1}), \forall l.\} \end{aligned}$$

The four proposed heuristics can be grouped into two categories: stochastic approximation and AI-based. Two stochastic approximation algorithms are myopic heuristic (MYO) and mean demand & supply heuristic (MDS); and two AI-based algorithms are myopic roll-out heuristic (MRO) and Iso-Net roll-out heuristic (IRO).

Myopic heuristic (MYO)

The MYO heuristic solves a single period problem in each decision epoch, which minimizes a single period cost given a system state $\mathbf{x}_t$.

$$\min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)] \quad (4.7)$$

It is named myopic as the cost-to-go function is ignored. (4.7) can be reformulated as a mix integer programming (MIP) problem.

Myopic roll-out heuristic (MRO)

The MRO heuristic use the myopic policy (calculated by MYO) as a base policy in the roll-out horizon. The cost-to-go function is approximated using parametric value approxi-

mation. We first rewrite (4.1)

$$\begin{aligned}
& \min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) + \beta v_{t+1}^*(\mathbf{x}_{t+1}(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{d}_t | \mathbf{x}_t))] \\
&= \min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)] + \beta \mathbb{E}[v_{t+1}^*(\mathbf{x}_{t+1}(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{d}_t | \mathbf{x}_t))] \\
&= \min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)] + \beta Q_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t),
\end{aligned}$$

where $Q_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t)$ is the Q-factor in epoch t regarding the state-action pair, i.e. $\{\mathbf{x}_t, \mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t\}$. Let $\phi(\mathbf{x}_t, \mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t)$ be a feature vector extracted from state and action information, and let θ_t be a parameter vector, we can linearly approximate the Q-factor as

$$Q_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t) \approx \theta_t^\top \phi(\mathbf{x}_t, \mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) = \hat{Q}_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t).$$

For convex approximations, let $\{\theta_t^j\}$ be a set of parameter vectors, we can approximate the Q-factor using convex piecewise linear function

$$Q_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t) \approx \max_j \{\theta_t^{j\top} \phi(\mathbf{x}_t, \mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t)\} = \hat{Q}_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t).$$

The feature space are selected to capture the most important characteristics associated with future costs. Feature extraction process is often handcrafted, based on human intelligence (see [44]) a simulator is used to collect data from which θ is trained. At epoch t , MRO first samples a vast number of actions. The simulator starts from state $\mathbf{x}_t$ and a sampled action, and run myopic policies for the following decision epochs. For each action, the simulator will collect several feature-cost pairs. A machine learning model (regression) is then implemented to obtain parameter vector(s) θ .

Then MRO solve an approximation of (4.1) which is a single stage stochastic optimiza-

tion problem

$$\min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)] + \beta \hat{Q}_t(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t | \mathbf{x}_t). \quad (4.8)$$

Mean demand and supply heuristic (MDS)

The MDS heuristic consider a two-stage optimization problem. The first stage cost is a single period cost, i.e. $C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)$, and the second stage is a projected cost in the future periods. We propose a mean demand and supply approximation, which essentially simplifies the probability structure of the cost-to-go function.

Let $\hat{\mathbf{d}}_{t+1}$ and $\hat{\mathbf{A}}_{t+1}$ be deterministic approximation of "future" demand and supply such that

$$\begin{aligned} \hat{d}_{t+1}^l &= \mathbb{E} \left[\sum_{i=t+1}^{\tau} \beta^{i-t-1} d_i^l \right] = \sum_{i=t+1}^{\tau} \beta^{i-t-1} \mathbb{E}[d_i] \\ \hat{A}_{t+1}^l &= \mathbb{E} \left[\sum_{i=t+1}^{\tau} \beta^{i-t-1} A_i^l | \mathbf{A}_t \right] = \sum_{i=t+1}^{\tau} \beta^{i-t-1} \mathbb{E}[A_i^l | \mathbf{A}_t]. \end{aligned}$$

To approximate state $\mathbf{x}_{t+1}$, we use mean demand, i.e. $\mathbb{E}[\mathbf{d}_t]$, to construct a deterministic state transition for any feasible actions, so that $s_{t+1}^l = s_t^l - m_t^l + \mathbb{E}[d_t^l] + w_t^l$. Note that we only use mean demand to approximate the number of specimens at the beginning of the second, but we still use the actual demand distribution to compute the costs in the first stage.

Therefore at each decision epoch, MDS solves the following problem

$$\min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t) \in \Pi(\mathbf{x}_t)} \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) + \beta \hat{v}_{t+1}(\mathbf{x}_{t+1})] \quad (4.9)$$

where the second stage is

$$\begin{aligned}
\hat{v}_{t+1}(\mathbf{x}_{t+1}) &= \min_{\mathbf{w}_{t+1}, \mathbf{a}_{t+1}, \mathbf{e}_{t+1}, \mathbf{q}_{t+1}, \mathbf{m}_{t+1}, \hat{\mathbf{d}}_{t+1} | \mathbf{x}_{t+1} \in \mathbb{R}^L} C(\mathbf{w}_{t+1}, \mathbf{a}_{t+1}, \mathbf{e}_{t+1}, \mathbf{q}_{t+1}, \mathbf{m}_{t+1}, \hat{\mathbf{d}}_{t+1} | \mathbf{x}_{t+1}) \quad (4.10) \\
s.t. \quad & 0 \leq \mathbf{a}_{t+1} \leq \hat{\mathbf{A}}_{t+1}, \\
& \sum_l w_{t+1}^l = 0, \sum_l e_{t+1}^l = 0, \sum_l q_{t+1}^l = 0, \\
& w_{t+1}^l \geq -(s_{t+1}^l - m_{t+1}^l + d_{t+1}^l) \quad \forall l, \\
& e_{t+1}^l \geq -(r_{t+1}^l - m_{t+1}^l + a_{t+1}^l) \quad \forall l, \\
& q_{t+1}^l \geq -(b_{t+1}^{l,0} - m_{t+1}^l + b_{t+1}^{l,1}) \quad \forall l.
\end{aligned}$$

We propose a cutting plane method to solve (4.9). The dual problem of (4.10) is can be derived

$$\begin{aligned}
\max_{\alpha, \gamma, \sigma, \eta \geq 0} \sum_{l \in [L]} & \left[\hat{A}_{t+1}^l \alpha^l + (m_{t+1}^l - r_{t+1}^l) \sigma_1^l + (m_{t+1}^l - b_{t+1}^{l,0} - b_{t+1}^{l,1}) \sigma_2^l + (m_{t+1}^l - s^l - \hat{d}_{t+1}^l) \sigma_3^l \right. \\
& + (r_{t+1}^l - s_{t+1}^l - \hat{d}_{t+1}^l) \eta_1^l + (s_{t+1}^l + \hat{d}_{t+1}^l - r_{t+1}^l) \eta_2^l + (b_{t+1}^{l,0} + b_{t+1}^{l,1} - s_{t+1}^l - \hat{d}_{t+1}^l) \eta_5^l \\
& \left. + (s_{t+1}^l + \hat{d}_{t+1}^l - b_{t+1}^{l,0} - b_{t+1}^{l,1}) \eta_6^l \right] \quad (4.11) \\
s.t. \quad & \alpha^l + \sigma_1^l - \eta_1^l + \eta_2^l \leq c_R, \quad \forall l \\
& \gamma_1 + \sigma_1^l - \eta_1^l + \eta_2^l - \eta_3^l + \eta_4^l = 0, \quad \forall l \\
& \gamma_2 + \sigma_2^l - \eta_5^l + \eta_6^l - \eta_7^l + \eta_8^l = 0, \quad \forall l \\
& \gamma_3 + \sigma_3^l - \eta_1^l + \eta_2^l - \eta_5^l + \eta_6^l - \eta_9^l + \eta_{10}^l = 0, \quad \forall l \\
& \eta_1^l \leq h_R, \quad \eta_2^l \leq p_R, \quad \eta_5^l \leq h_B, \quad \eta_6^l \leq p_B, \quad \forall l \\
& \eta_3^l + \eta_4^l = \frac{K_R}{2}, \quad \eta_7^l + \eta_8^l = \frac{K_B}{2}, \quad \eta_9^l + \eta_{10}^l = \frac{K_S}{2}. \quad \forall l
\end{aligned}$$

For a given first stage action, the second stage state is obtained and substituted in the dual problem (4.11). Solving the dual problem produces a set of optimal dual variables

$(\hat{\alpha}, \hat{\gamma}, \hat{\sigma}, \hat{\eta})$. Let $\hat{V} = \hat{v}_{t+1}(\mathbf{x}_{t+1})$, a cutting plane is added in the the first stage problem

$$\begin{aligned} \hat{V} \leq \sum_{l \in [L]} & \left[\hat{A}_{t+1}^l \hat{\alpha}^l + (m_{t+1}^l - r_{t+1}^l) \hat{\sigma}_1^l + (m_{t+1}^l - b_{t+1}^{l,0} - b_{t+1}^{l,1}) \hat{\sigma}_2^l + (m_{t+1}^l - s^l - \hat{d}_{t+1}^l) \hat{\sigma}_3^l \right. \\ & + (r_{t+1}^l - s_{t+1}^l - \hat{d}_{t+1}^l) \hat{\eta}_1^l + (s_{t+1}^l + \hat{d}_{t+1}^l - r_{t+1}^l) \hat{\eta}_2^l + (b_{t+1}^{l,0} + b_{t+1}^{l,1} - s_{t+1}^l - \hat{d}_{t+1}^l) \hat{\eta}_5^l \\ & \left. + (s_{t+1}^l + \hat{d}_{t+1}^l - b_{t+1}^{l,0} - b_{t+1}^{l,1}) \hat{\eta}_6^l \right] \end{aligned}$$

The cutting plane method starts from solving a myopic single period problem to generate the first cut. Then more cuts are added iteratively and (4.12) is solved repeatedly until no significant improvement.

$$\begin{aligned} \min_{(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t)} & \mathbb{E}[C(\mathbf{w}_t, \mathbf{a}_t, \mathbf{e}_t, \mathbf{q}_t, \mathbf{m}_t, \mathbf{d}_t | \mathbf{x}_t) + \beta \hat{V}] \\ \text{s.t.} & \quad \hat{V} \leq \text{cuts generated by (4.11)} \end{aligned} \quad (4.12)$$

Iso-Net roll-out heuristic (IRO)

The IRO heuristic assumes that transshipment/relocation are not allowed in the following decision epochs which uses the Iso-Net policy as a base policy in the roll-out horizon. The only difference between IRO and MRO is that when collect feature-cost data using the simulator, the myopic policies are replaced by Iso-Net policies.

4.6 Dynamic Resilience and Cost Efficiency

We investigate dynamic resilient network design (Co-Net) and its cost efficiency comparing to a traditional resilient network design (Iso-Net). Consider a decentralized network with 2 cell therapy manufacturing facilities (i.e. $L = 2$), and we adopt the supplier disruption setting in the case study of Chapter 3. We assume facility specifications are identical (see Table 4.1), but their suppliers may have independent or correlated disruption processes.